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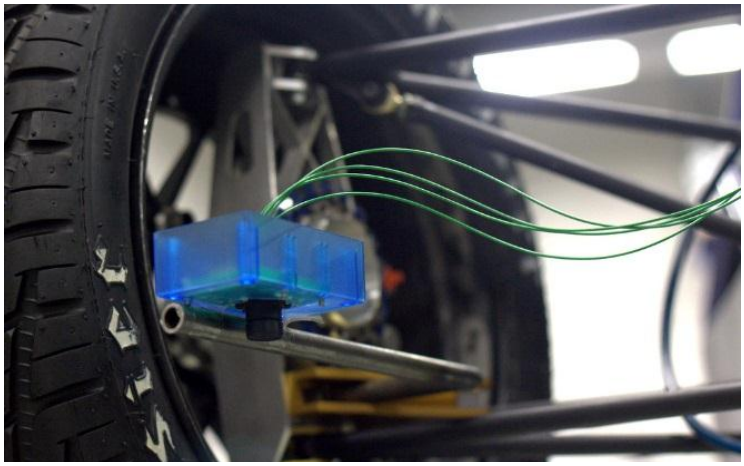
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Development of a Low Cost Slip-Angle Sensor



Slip Angle Sensor Mounted on the 2013 University of Wollongong F-SAE Vehicle

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ABSTRACT

For all road vehicles, as tyres are the only point of contact with the ground, all dynamic forces acting on the vehicle must be reacted through the tyres. Tyres generate forces through the deformation of their surface in contact with the ground. This deformation of the tyre is known as slip. Two different parameters for slip can be measured: slip ratio, the difference between the speed of the driving tyres and the ground speed, and slip angle, the angle between the pointed direction and direction of travel of the tyre.

Currently, slip angles are difficult to measure, requiring either indirect measurement methods or expensive and limited specialised sensors. This paper details the use of an optical sensor from a gaming mouse as a cheap, yet accurate alternative, increasing potential for the measurement and analysis of tyre slip angles.

The developed sensor package is able to measure slip angles, with a mean angle error of 0.05^0 and standard deviation of 0.11^0 , under laboratory testing conditions. It is also able to measure slip angles while accounting for a large range of environmental and packaging variables.

INTRODUCTION

This paper summarises the work done as part of a project, with the aim of developing and implementing a slip angle sensor for a Formula SAE style race car.

Formula SAE

Formula SAE (FSAE) is an international design competition for University students. Teams of students are tasked to conceive, design, fabricate, develop and compete with small, formula style, vehicles [1]. The competition started in the USA in 1981, with 6 competing teams. Today there are hundreds of teams, with official competitions held in USA, Australia, Brazil, Italy, UK, Austria, Germany and Japan.

A competitive car and good design requires not only that the students in the project to have a strong understanding of the theoretical concepts that explain how the vehicle behaves, but also that they possess the ability to collect data from current vehicles and use this real-life information to tune the car and correct or confirm any assumptions made in the theoretical model [2]. The latter part requires that the vehicle be equipped with an array of sensors to accurately log important parameters for later analysis of the vehicle's performance. Beneficial to vehicle tuning and performance are the sensors that log suspension system parameters, the system which ultimately controls the behaviour of the tyres.

Slip and Slip Angles

For most vehicles, the only part of the vehicle that is in contact with the ground is the tyres. Therefore, all vehicle handling characteristics and dynamic capabilities must exist due to the forces generated and reacted through the tyres [3]. The forces generated by the tyre are a result of the elastic deformation of the tyres surface when interacting with the road surface. This characteristic is known as slip. In the case of lateral forces, the tyre surface deforms normal to the centreline of the tyre itself, as shown in Figure 1.

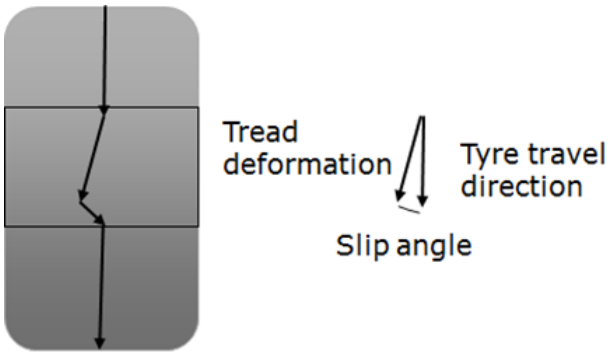


Figure 1: Deformation of the tyre contact patch

Drawing a line from the initial position of an element of tyre material to the point of maximum lateral deflection, and measuring this against the tyre centreline heading gives the slip angle [4].

The slip angle is the actual tyre heading relative to the tyre direction defined by the alignment of the tyre tread at the road-tyre interface. As long as the vehicle isn't in an undisturbed straight line, the slip angle will not be zero

All tyres have different characteristics with regard to the way they produce forces given varying slip angles due

largely to the difference in tyre compounds. Figure 2 shows the relationship between normal force and slip angle.

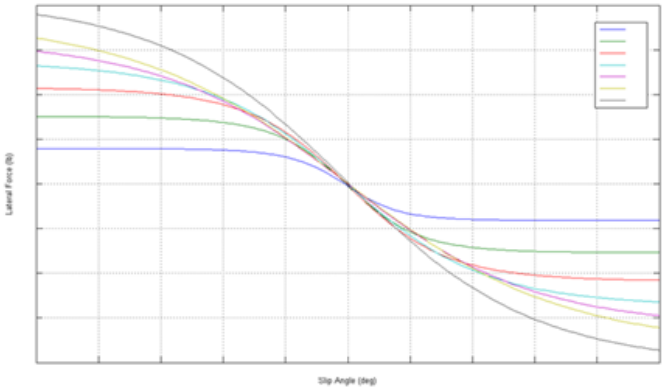


Figure 2: Relationship between lateral force and slip angle for different normal loads

It can be seen that as slip angle increases for a given normal force, there is also an increase in lateral force, which typically plateaus or drops off at very high slip angles.

Measuring Slip Angles

The measurement of slip angles has been limited to date, with the majority of measurement occurring in laboratory facilities, involving only small amounts of data and high levels of secrecy, typically for motorsport applications. This has been due to the lack of advanced technology to accurately measure the tyre slip, especially in a packaging format suitable for racing, where weight is a key priority. Where this technology has started to become available, cost has become a limiting factor. The development of a cheap and accurate sensor also serves a need outside of motorsport, with car modifications, emerging manufacturers, private builders, private research and development initiatives and education who could use the data in the development, but may not have sufficient budget to make the acquisition of the sensor feasible. Production cars could also benefit from the technology, with future development possibly leading to an increased range of standard vehicle sensors available to customers, allowing for more thorough stability controls.

Currently there are a couple of different ways to measure slip, both directly and indirectly. These methods include the use of optical sensors or GPS fused with other on-board sensors.

GPS systems have improved over the years, in terms of both price and accuracy. With two GPS receivers (one mounted on the front of the vehicle, and the other on the rear) it is possible to determine the true heading of the vehicle at any time, and then translate this information out to the positions of the axles and individual tyres, and calculate the slip angles there. Advancements in GPS technology, such as DGPS, which uses a base station placed in close proximity, can reduce error from the transmission through the atmosphere compared to satellite technology. Combined with higher update rate receivers and other standard suspension sensors such as gyros, accelerometers and shock pots, give a good picture of how the vehicle is moving. However, the use of GPS is still limited by the ultimate positioning accuracy of the receivers, which is in the order of 10 cm at best. The update rates are still limited to about 100Hz, and the system directly reads slip of the vehicle as a whole, it then uses math, geometry and the other sensors to then determine the values of tyre slip.

Optical sensors are one way of measuring tyre slip directly, rather than through a combination of math and kinematics. These types of sensors work by taking an image of the ground below them and comparing that to the next image taken, with feature recognition algorithms identifying notable features and tracking their movement across the frame. This makes it possible to calculate the distance in the x and y direction that the sensor has moved. To do this requires some considerable computing power by the sensor, along with a high frame rate. If the two sequential images do not overlap by 50%, an accurate recognition of features and distance travelled can't be calculated. This method of sensing slip is more accurate due to its direct measuring of the parameter, however due to frame rates and processing power required by the chip, they are very expensive to purchase. The industry standard sensor, the Kistler Correvit SFII-P used by high level motorsports teams such as those in F1, is cost inaccessible to teams with lower budgets and FSAE teams.

SENSOR TECHNOLOGY

Optical mice work on the same principal as the optical sensors mentioned in the previous section, with the only exception being the shorter lens focal length than that of the SFII, with mice being designed to work at a distance of 2mm from the surface compared to 180mm.

Over the last decade, mouse technology has advanced significantly, with the gaming market, in particular moving towards cheaper and more accurate gaming mice. Current gaming mice are therefore the ideal candidates for the use in this project. Choice of sensor from an optical mouse will be determined by comparing 4 major factors. These are frame rate, image size, resolution and cost.

Frame rate refers to the number of images that the sensor is capable of taking per second. This parameter determines how quickly the sensor is able to take successive images of the surface, and therefore a sensor with a higher frame rate is able to track features on the ground at higher speeds, since less time passes between frames.

Image size refers to the number of pixels that make up each image. Higher image sizes allow a larger image to be captured by the sensor and therefore allow the sensor to track the ground at a higher speed.

Resolution refers to the level of surface details that the sensor is able to detect. Resolution is measured in counts per

inch (cpi). The higher the value of resolution the more detail on the surface the sensor is able to detect. This allows the sensor to move more quickly across the surface, at a higher accuracy than a lower resolution sensor.

Cost refers purely to how much the sensor costs to purchase. As the aim is to make the entire unit as cheap as possible. However, the popularity of gaming mice means that they are all relatively cheap, with the more expensive mice costing only $100.

The performance aims for this project were to make the sensor as accurate as possible, and to allow for readings to be useful and consistent when analysed post-test. To achieve this, values for frame rate, image size and resolution of the sensor have to be as high as possible to allow for accurate readings over the speed range seen in FSAE, and as an extension other forms of motorsport. Table 1 details the performance specifications of the SFII, to which the performance of this sensor will be compared.

Table 1: SFII performance specifications

Speed range	0.3 – 250 km/h
Distance resolution	2.08 mm
Uncertainty of measurement	< ± 0.5%
Speed linearity – desired distance	< ± 0.5%
Distance linearity	< ± 0.2%
Working distance and range	180 ± 50 mm
Angle range	± 40 degrees
Angle resolution	± 0.1 degrees

The specifications that were compared for this sensor are the values of resolution and accuracy of angle, with some emphasis on the speed range as well. A loan of a SFII has been arranged to verify these values and compare the sensors side by side under the same conditions.

Table 2: Comparison of mouse sensors

Sensor		*ADNS 2610*	*ADNS 2051*	*ADNS 3055*
Image size	*Pixels*	18x18	18x18	30x30
Frame rate	*fps*	1500	500 – 2300	500 – 6666
Resolution	*cpi*	400	400 – 800	400 – 2500

Table 2 compares three of different optical sensors that are currently on the market. The ADNS-2610 and ADNS-2051 are typical of the type of sensor found in most desktop mice, with a comparatively low frame rate and resolution. The ADNS-3055 has one of the highest frame rates and resolution currently available. These sensors are widely available and low cost, with both the sensor and mouse body costing approximately $30. Due to its high frame rate and resolution, the ADNS-3055 was selected out of the initial 3 considered as the sensor upon which to base the developed tyre slip angle sensor.

The standard packaging and lens for the ADNS-3055 allow for tracking of movements up to 4.06m/s. A new lens was added to allow the board for the sensor to be mounted further away from the ground, which reduced the likelihood of impacts with objects on the road and allowed for a larger imaging area based on the focal length of the lens.

An M12 standard mount lense~~s~~ was used to allow for flexibility in focal length.

Communication

The sensor stores data about various operating conditions in on-board registers [5]. The information stored in these registers include product ID, delta x and y movement, shutter speed, surface quality and others. These registers can be read/written to via the SPI protocol.

All vehicles manufactured since 2001 implement a form of CAN-bus, particularly in motorsport, where most ECU/dataloggers implement the bus network for communicating large amounts of data to and from many different modules. Original development of the CAN-bus started in 1983 by Bosch, and was officially released in 1986 [6]. The CAN-bus network is a message-based

protocol, with a message consisting of an ID, followed by the data bytes. In the case of the University of Wollongong FSAE team, data logging is carried out by a MoTeC M800 and ADL3, both of which can accept CAN-bus inputs.

A module was designed and built to allow for the communication between the ADNS-3055 registers and the vehicles CAN-bus, as laid out in figure 3. The board features an ATMega328 processor, with an MCP2515 CAN controller and MCP2551 CAN transceiver. This board allows for code to be written to access the sensors Δx (delta_X) and Δy (delta_Y) registers, where the movement data is kept, and this data sent on to the CAN bus, for later analysis in i2 Pro. During the development of the sensor, this code was written to read the SQUAL (Surface Quality) register, for analysis of how the sensor was affected under different operating conditions.

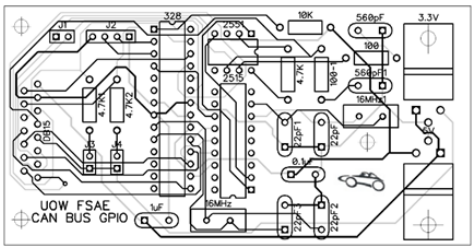


Figure 3: CAN-Bus GPIO board layout

The board was designed to not only communicate data from the slip angle sensor, but with different code to also work as a general input/output device, with 6 analogue inputs, 4 digital inputs/outputs and serial bus, as well as 5V and 3.3V regulators.

A printed circuit board was also developed for the sensor, including mounting holes for the sensor and lens, ensuring correct orientation and alignment, refer to Figure 4.

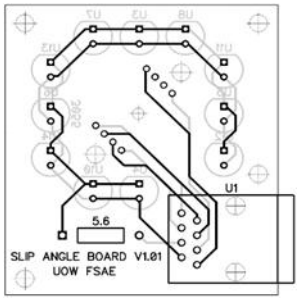


Figure 4: Sensor circuit board

TESTING

To allow for the testing and characterisation of the sensor, different rigs were manufactured.

The main test rig was built to simulate various angles and compare these to the values read by the sensor. A large metal plate was machined with 31 holes centred around a common hole distanced 250mm away in order to accurately set the sensor at angles from -30° to +30° in 2° increments, as shown in figure 5. The angle test rig was positioned above a conveyor belt, which acted as amoving road. Once positioned above the conveyor, the selected angle could be easily changed, allowing for comparison between the set angle and angle measured by the sensor, which was pointed at the moving belt.



Figure 5: Angle test rig

Other mounting rigs were manufactured to test additional characteristics of the sensor, as needed. A mount was built for a height adjustable trolley to compare the effects of packaging on the sensor. Another rig was built which attached to the tow-ball of a car, as shown in figure 6, to test the effect of speed and check the operation of the sensor in a dynamic environment.



Figure 6: Car test setup

Initial Testing

Due to the intended use of the sensor, in a gaming mouse application, the ADNS-3055 has inbuilt correction to allow for more smooth linear movement along its x and y axis, ignoring small fluctuations in movement which may occur. This allows for straight movement of the cursor when used as a mouse, but interferes with the readings made by the tyre slip angle sensor.

The sensor was placed on the conveyor rig, with the axis of the sensor aligned with that of the belt (X lengthways, Y

perpendicular). The graph in Figure 7 shows the result of the test.

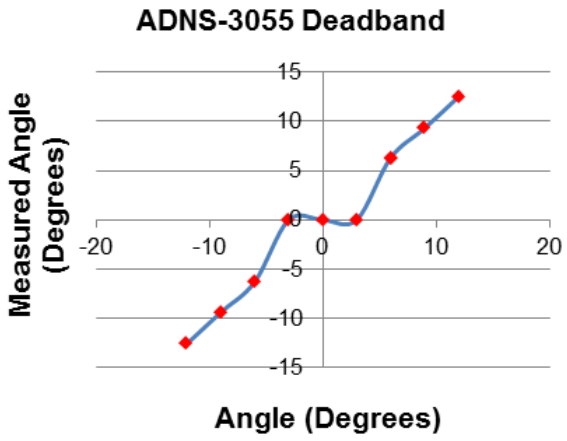


Figure 7: Tested sensor deadband

It was confirmed that the sensor does have a deadband from approximately -5° to 5°. As this is a large range relative to the measurement of slip angles this region had to be avoided. Tyre data on common tyres previously used for FSAE shows that, dependant on the normal load, 5° may be where the peak lateral force occurs. If the sensor is unable to determine the angles up to and including 5°, little valuable data will be provided under normal operating conditions.

The sensor was rotated 45°, realigning the x and y axis and avoiding interaction with the deadband in normal operation, allowing the sensor to constantly measure small slip angles around 0° with accuracy. The only negative consequence of this arrangement was that at speeds slightly above zero, the sensor would not operate correctly as some component of the sensor's direction of travel aligned with the original x and y direction. However, this is a minor issue as the range is very small, and the vehicle is very rarely travelling at such slow speeds.

Height

During testing to evaluate the sensors ability to measure over a range of changing environmental and packaging variables, the value of surface quality read by the sensor was measured and compared. Surface quality is a measure of the number of uniquely identifiable features that the sensor is able to see in the current frame, which it then uses for movement tracking. The higher the value, the more objects the sensor is able to track and more precisely calculate the movement of the sensor, frame to frame.

The manufacturer of the sensor supplies surface quality data, applicable to tests using their recommended settings (lens focal length, height, illumination), where the sensor is moved slowly across a piece of white paper. The average SQUAL value, from the data shown in Figure 8, is approximately 32 counts. This value was used as a baseline.

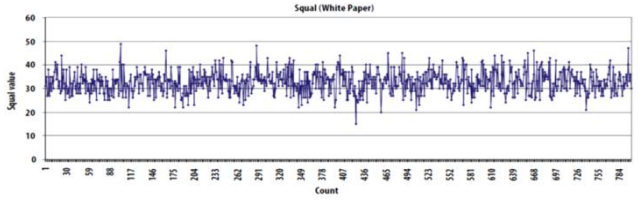


Figure 8: Baseline surface quality values

A vehicle moves up and down vertically due to the change in road gradient and vehicle dynamics, such as weight transfer. The sensor has to be able to deal with these changes in distance between it and the road surface. Having a large vertical operation range allows for a freedom in packaging on the vehicle, making it easier to mount if a tight tolerance on height has to be met.

The sensor package was mounted to a height adjustable trolley, placed over a road surface, and the height from the lens to the ground increased. Figure 9 shows the value of surface quality compared to height, with the baseline (32) from the manufactures test also plotted.

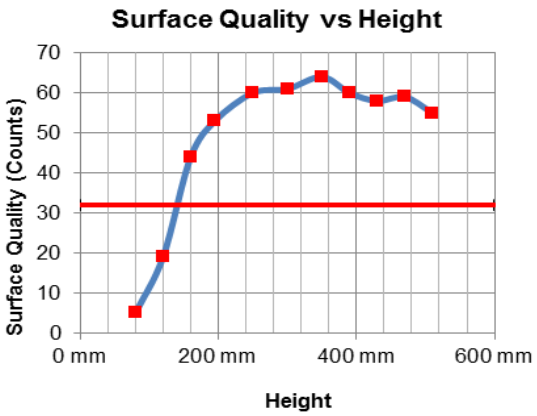


Figure 9: Surface quality vs height test

From Figure 9 it can be seen that there is a minimum focusing distance. At distances closer than this minimum, the sensor is unable to focus clearly on the ground and pick up features on its surface. At a distance of 140mm between the surface and lens, the sensor crosses the benchmark of 32 counts and continues to rise until approximately 250mm, where it levels off. Between 250mm to 500mm the surface quality remains roughly the same, only starting to drop off slightly from 460mm to 510mm, after which the test was stopped.

The results of this test indicate, as shown in Figure 9, that the sensor can be mounted on the vehicle from 240mm to 500mm, with only a small change in performance in terms of its ability to measure features on the road surface. If packaging constraints required it, it could be taken down to approximately 150mm and beyond 510mm and still be able

to effectively measure tyre slip angles. Such a large operating range allows for flexibility in the placement and operation of the sensor.

Illumination

Being an optical sensor, light is needed for the sensor to operate. Without the correct illumination, the sensor is unable to see the surface features, and therefore, unable to track them and calculate movement. Just like a camera, the sensor is able to automatically adjust its shutter open time, to an extent, to accommodate for the amount of light being captured. If too little light is available, the sensor will no longer be able to accommodate and the resulting images captured will be progressively darker, making the identification of surface features more difficult to achieve.

The sensor was tested over a range of heights, first in bright sunlight, then in the shade on the same patch of the ground, as the sun set. Using a light meter, the bright sun measured 25 Lm, and the shade measured 10 Lm. Figure 10 shows the results of this test.

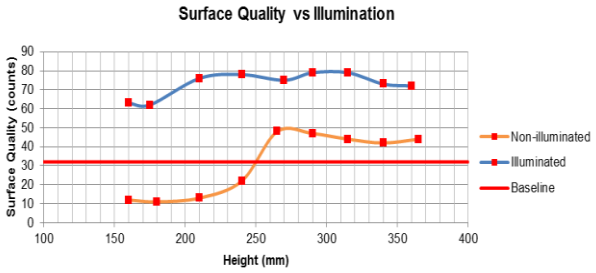


Figure 10: Surface quality vs illumination test

The tests with the illuminated surface show that the sensor is able to constantly identify substantially more surface features for tracking, while the non-illuminated test exceeds the baseline after height of 250mm.

As the environmental light is not always constant, 11 high brightness LEDs were added to the final senor board to allow a constant suitable level of light to always be present on the section of road that the sensor is measuring.

Speed

For the sensor to work effectively, an overlap of at least 50%, frame to frame, needs to be maintained in order for it to be able to track the movements of enough features on the surface. The sensor was attached to a vehicle in order to test the sensors ability to operate at speed, in a dynamic environment. Figure 11 shows the results of this test.

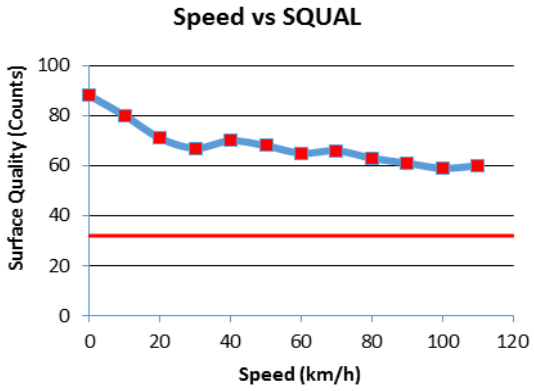


Figure 11: Surface quality vs speed test

The sensor is quite capable of distinguishing surface features up to 110km/h (the national highway limit). As expected, at high speed there is a decrease in total counts identifiable by the sensor, due to the overlap becoming larger and larger. However, the decline is relatively low and even at 110km/h, is still almost double that of the baseline for the sensor. In the case of FSAE vehicles, mid corner speeds, (where slip angles data is most valuable) are relatively slow at around 50km/h. Maximum speed at the end of a straight is approximately 105km/h.

RESULTS

Figure 12 shows the results of the testing of the sensor.

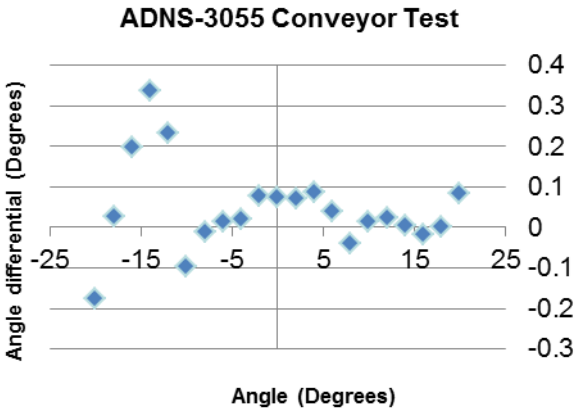


Figure 12: Total sensor accuracy

Figure 12 shows that the final sensing unit can measure slip angles very accurately. For the data above, the mean error is 0.05° with a standard deviation of 0.11°. For comparison, the accuracy of the SFII from Kistler, which is used by vehicle dynamics companies and high levels of motorsport, has a mean error of 0.1°. This comparison shows that the

results achieved by the sensor are at least on par with the leading commercial unit available.

Final Product

The final product is clearly able to accurately measure slip angles, over a wide range of environmental variables. The unit produced by this project contains custom printed circuit boards to transmit power and data between the chips and datalogging devices. They are all housed in plastic containers to weatherproof them from the environment, with standoffs for the printed circuit boards for vibration isolation.

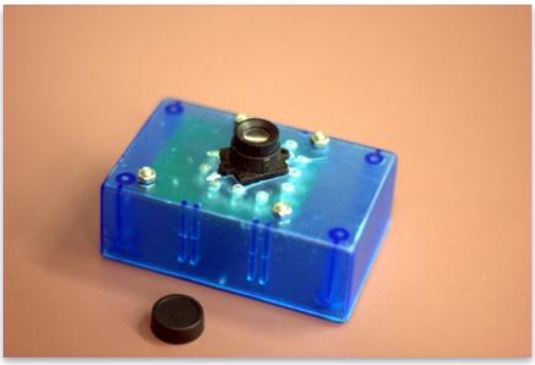


Figure 13: Final product

The cost of the sensor unit is approximately $25, including the optical sensor, lens, LEDs, printed circuit boards and case. The communications board costs around $50 including the processor, CAN controller and transceiver, voltage regulators, passive components and connectors. The complete unit can be made for less than $100.

CONCLUSION

A sensor package was developed that can accurately measure slip angles (0.05° mean angle error). It is able to perform under a large variation in environmental factors such as the packaging height, illumination and speed. It is also able to do this for a total cost of under $100, allowing for the technology to be available to groups that would not have had the opportunity to use it before.

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DEFINITIONS/ABBREVIATIONS

CAN	Controller Area Network
ECU	Electronic Control Unit
GPIO	General Purpose Input/Output

PUBLISHED REVIEWS

To assist readers and expand the research community, reviewers publish a review of the final paper. Enter no text here and do not delete this section. This will be managed by the editorial team later.

Shane Richardson
Delta-V Experts

This paper represents an innovative and pragmatic application of repurposing and development engineering, which achieves a application specific outcome at a significantly lower price point whilst maintaining high measurement accuracy. Each Formula SAE team should be either contacting Nathan and/or developing their own system based on the approach and data presented. Anyone undertaking vehicle testing or development which requires an understanding of tyre slip angles should seriously consider reading, absorbing and/or developing a system based on this paper.

Clint Steele
Swinburne University of Technology

This paper has shown an example of cost effective methods of measuring vehicle performance. The author is correct in pointing out that this is of value to both typical student race teams and small auto companies, timely given the current shift in the local auto industry. However, it also shows what can be done in the area of educational equipment development as well. Not to mention small race teams generally. Finally, it is also a good reminder of how one need not rely on large budgets for insightful engineering analysis. It would be good to see further work into the development of other cost effective sensors and economical enhancements of such sensors for improved vehicle tuning published. It would be a benefit to many researchers as is, but also useful to understand what kind of R&D budgets really are needed for automotive R&D.